

Bryanna Downing

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EDUCATION

University of Florida

Gainesville, FL

B.S. in Biomedical Engineering

May 2027

Minor in Bioinformatics

Achievements:

- College of Engineering Dean's List, Fall 2024 – Present; University Scholars Program; Bank of America Scholar; Arthrex Scholar

Relevant Coursework:

- Intro to Bioinformatics, Python Programming for Biologists, Cellular Engineering Lab, The Microbiome, Quantitative Physiology

Broward College (Dual Enrollment)

Davie, FL

Associate of Arts

August 2022 – May 2023

Achievements:

- Highest Honors; Academic GPA: 4.0/4.0

EXPERIENCE

Undergraduate Research Assistant

September 2025 – Present

University of Florida

Porras TMI Lab, Gainesville, FL

- Evaluating dbCAN and Cayman for carbohydrate-active enzyme annotation across *Bacteroides fragilis* genomes and related type strains.
- Comparing CAZyme family annotations, substrate coverage, and substrate specificity with emphasis on host-derived glycans and extracellular matrix components.
- Performed BLAST searches using *bft-1*, *bft-2*, and *bft-3* reference genes to examine genomic variation across bacterial strains.
- Utilized **R** and **Python** to visualize distribution of genomes across different hosts and disease states.

Teaching Assistant – Cellular Systems and Physiology

January 2026 – May 2026

University of Florida

BME, Gainesville, FL

- Mentored a small group of 9 students through weekly lab sessions, enhancing their understanding of human physiology through engineering principles and fostering independent and collaborative problem-solving.

Sales Associate

January 2023 – August 2023

Tillys

Coral Springs, FL

- Exceeded monthly sales targets by matching customer needs with the appropriate clothing, staying current with the evolving fashion trends.
- Processed customer transactions quickly and accurately while supporting efficient checkout and return operation

PROJECTS

Fibroblast Proliferation on a Collagen-Modified Surface

Cellular Engineering Lab

- Examined the effect of Type I collagen-coated surfaces on fibroblast proliferation, cellular behavior, and metabolism
- Assessed cell viability, morphology, metabolic activity, and wound healing abilities through assays such as Live/Dead & Hoescht staining, Alamar Blue, brightfield microscopy, and scratch assays.

SKILLS & CERTIFICATIONS

Programming: Python, MATLAB, R, Bash, Excel, HTML, CSS

Bioinformatics: BLAST, dbCAN, Cayman

Tools: Git, GitHub, LaTeX

Certifications: Certified SolidWorks Associate (CSWA)